Laksh Goyal

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Education

University of California San Diego GPA: 3.6
B.S. in Computer Science; Cognitive Science and Business Economics minors *Expected Graduation: Dec 2027*
Relevant Coursework: Algorithms & Data Structures; Operating Systems; Computer Architecture

Achievements

15x Hackathon Winner: 1st Gemini, 1st GitHub Copilot, 1st Replay (AWS), 1st Akash Cloud, 1st Convex
2x Founding Engineer: Pylon ($100K raised, Bow Capital incubator); Inovasity (3D wound imaging and analysis)
Open Source Contributor: CERN Labs - Awkward Array [https://github.com/scikit-hep/awkward]

Experience

DR Financial Remote
Software Engineer Intern *Jul 2026 – Present*
- Built DR Financial's first product to check doctors' licenses and credentials, reducing a 4-hour manual process to 2 minutes of automated checks plus 10 minutes of human review.
- Caught 28 of 29 seeded credentialing defects with offline regression tests on every merge, without live lookups.
- Caught every excluded-doctor match in 1,500 adversarial test cases against the 83,665-row federal exclusion list.
- Shipped the credentialing service on AWS ECS/Fargate, using ECR for Docker images and S3 for reviewer evidence.

GMI Cloud (**NVIDIA** Reference Platform Cloud Partner) Mountain View, CA
Software Engineer Intern *Jun 2026 – Sep 2026*
- Built Mint, a React/FastAPI lead-review platform still used internally at GMI Cloud, cutting each batch from 20 staff-hours to 20 minutes of review by 1 person.
- Onboarded Mint's 20-person Growth, Events and Marketing team in 50 minutes with docs and walkthroughs.
- Separately, built a coding-model router that escalates checker-rejected answers, solving 94/100 frozen EvalPlus tasks at 47% lower cost than the published single-model baseline. Load-tested 500+ requests.

Qualcomm Institute San Diego, CA
Machine Learning Researcher *Oct 2025 – Jun 2026*
- Built Python pipelines for EfficientNet wildlife-audio research, increasing training-ready recordings from 7% to 15% of raw archives through metadata joins and corrupt-file filtering.
- Raised EfficientNet training throughput 30% with prefetching and parallel data loading, reducing GPU time waiting for data from 31% to 10% in measured wildlife-audio runs.

AWS Student Builder Club @ UC San Diego San Diego, CA
Tutor and Technical Workshop Lead *Sep 2026 – Present*
- Tutored 20 students in system architecture, databases and deployment through hands-on AWS workshops, using worked examples and guided troubleshooting to help members build and deploy their projects.

Projects

Leaf: **macOS File Search** | Swift, SwiftUI, AppKit, Spotlight Demo | GitHub
- Built Leaf, a native macOS file-search app with 100 installs and 30 weekly users, handling about 200 searches per day and previewing PDFs at matching pages.
- Combined cancellable Spotlight queries with direct folder scans to find new downloads before indexing, adding typo-tolerant ranking and byte-verified duplicate detection.

Hacklist: **Hackathon Calendars** | JavaScript, Cloudflare Workers Live | GitHub
- Built Hacklist, auto-updating hackathon calendars with 100+ subscribers each in San Francisco and San Diego.
- Raised signups 3.4× after user feedback by embedding Luma RSVPs directly in event-discovery cards.

Basis: **Brokerage Account Checker** | Java, Spring Boot, PostgreSQL Live | GitHub
- Built Basis, a Java double-entry ledger for 90 real accounts across 10 brokers in UCSD's investment-group network, explaining 14 cost-basis errors from source trades.
- Prevented concurrent ledger updates from consuming the same shares twice with a PostgreSQL advisory lock, verified by matched race tests against an unlocked control.

Technical Skills

Languages: Python, C++20, Java 21, Go, Rust, C, TypeScript, JavaScript, SQL, Bash, HTML/CSS, Swift, Kotlin, R
Frameworks & APIs: SwiftUI, AppKit, React, Node.js, FastAPI, Spring Boot, Flask, PyTorch, TensorFlow, REST APIs, gRPC
Data & Infrastructure: PostgreSQL, Redis, SQLite, Docker, Kubernetes, AWS EKS, EC2, Lambda, S3, IAM, Terraform
Systems & Tooling: Linux, Git, GitHub Actions, CI/CD, CMake, Gradle, pytest, JUnit 5, Playwright, libFuzzer, Prometheus